Acta Materialia 176 (2019) 19–32

Contents lists available at ScienceDirect

Acta Materialia

journal homepage: www.elsevier.com/locate/actamat

Full length article

Oxygen effects on crystal plasticity of Titanium: A multiscale calibration and validation framework

C. Kale, P. Garg, B. Gholami Bazehhour, S. Srinivasan, M.A. Bhatia, P. Peralta, K.N. Solanki*

School for Engineering of Matter, Transport, and Energy, Arizona State University, Tempe, AZ, 85287, USA

ARTICLE INFO

Article history:
Received 21 February 2019
Received in revised form
7 June 2019
Accepted 24 June 2019
Available online 29 June 2019

Keywords:
Oxygen
Titanium
Crystal plasticity
Density functional theory

ABSTRACT

This work investigates macroscopic plastic behavior in polycrystalline Ti with varying oxygen concentrations using both experiments and a crystal plasticity finite element framework applied to a representative volume element (RVE) of the microstructure. The proposed multiscale framework makes use of parameters such as critical resolved shear stress (CRSS) ratios obtained from lower length scale first-principle calculations and strain hardening rates obtained from experiments. Generalized stacking fault energy calculations (GSFE) in combination with the Peierls-Nabarro model and a temperature dependent phenomenological description of CRSS were utilized to compute the CRSS ratios for different slip systems in Ti with and without oxygen. Experimentally measured stress-strain responses along the rolling direction for polycrystalline Ti with different oxygen concentrations were used to obtain the strain hardening rates. The crystal plasticity framework was then calibrated using the computed CRSS ratios and the strain-hardening rate for Ti with various oxygen concentrations. The calibrated model was then used to predict the macroscopic response of Ti under different loading conditions and orientations with different oxygen concentrations. Without tuning the fitting parameters for the crystal plasticity framework, we show that the model is able to simulate the experimental responses within the experimental uncertainty. Thus, a systematic calibration procedure is presented to capture the macroscopic homogeneous responses of Ti with various oxygen concentrations.

© 2019 Acta Materialia Inc. Published by Elsevier Ltd. All rights reserved.

1. Introduction

Titanium (Ti) and its alloys have been traditionally used for structural applications in automotive, aerospace, and biomedical applications due to their high strength-to-weight ratio [1,2]. Commercially pure (CP) and extra low interstitial (ELI) Ti-alloys, in particular, are attractive materials due to their excellent corrosion resistance, lightweight, and formability. An important factor with significant effects on the macroscale mechanical properties of these alloys needed for the above applications is the activation of various slip systems. For instance, like other hexagonal close packed (HCP) structures, Ti has five commonly observed slip systems with prismatic ($\{10\bar{1}0\}1\bar{2}10$) being the principal slip system [3]. Hence, characterizing the slip resistance of each system, i.e., the critical resolved shear stress (CRSS) at which dislocation slip starts, is crucial in understanding the anisotropic plastic deformation in Ti [4–6]. However, assessing CRSS values through experiments using either single crystals (direct measurements, which generally involve multiple slip systems or deformation modes), or large polycrystalline samples (indirect measurements) have led to large variations in the reported CRSS values, see Wang et al. [7] and also refer to Table 1. For instance, Conrad compiled CRSS values for the prismatic and basal slips and reported a basal to prismatic CRSS ratio of ~3–4 for high purity Ti at room temperature [8]. Conversely, Barkia et al. [9], used slip trace analysis in polycrystalline samples under tension and reported basal to prismatic, and $\pi 1$ <c+a> to prismatic CRSS ratios for two different grades of CP-Ti to be roughly 1.5 and 2, respectively. More recently, Wang et al. [7] analyzed the grains in polycrystalline samples individually and found values of CRSS ratios for basal to prismatic slip to vary from 1.7 to 2.1 for CP-Ti grade 1. Such divergent experimental results point to the need for a systematic methodology to determine the anisotropic plastic behavior of Ti alloys. Moreover, the large variation in the reported CRSS values from experiments has been attributed to variable solute contents, testing methods, strain rate, texture, etc. [7,9–11].

On the other hand, continuum-mesoscale polycrystalline simulations (with various fitting parameters) are commonly used to

* Corresponding author.
E-mail address: kiran.solanki@asu.edu (K.N. Solanki).

https://doi.org/10.1016/j.actamat.2019.06.040
1359-6454/© 2019 Acta Materialia Inc. Published by Elsevier Ltd. All rights reserved.

Table 1
Range of scatter in the normalized CRSS data (normalized with respect to the prismatic slip system in each case) in the literature for dominant slip systems in pure Ti.

Material	O (wt. %)	Basal	π_1 <a>	π_1<c+a>	π_2<c+a>	Reference
High purity Ti	0.009	1.3	N/A	5.3	N/A	[14]
CP-Ti grade 1	0.07	1.2	N/A	2.6	N/A	[19]
CP-Ti T40	0.12	7	4	8	5.9	[12]
CP-Ti T40	0.16	1.5	1.2	2	N/A	[9]
CP-Ti grade 1	0.17	1.7–2.1	N/A	≥2.4	N/A	[7]
CP-Ti grade 2	0.25	3.6	25.3	22.5	N/A	[18]
CP-Ti grade 2	-	2.3	N/A	7.4	N/A	[16]

determine the CRSS values for different slip systems [12–16]. To achieve that goal, the parameters of the chosen model are determined by insuring that the simulated global response of an appropriate microstructural realization shows an acceptable level of agreement with experimental results. Often, macroscopic stress-strain curves are used as references for model calibrations. For instance, Salem et al. used a visco-plastic Taylor-type crystal plasticity constitutive law and calibrated its parameters using uniaxial compression experiments and reported the basal to prismatic, and π_1 <c+a> to prismatic CRSS ratios to be 1.3 and 5.3, respectively [14]. Mayeur and McDowell presented a simple procedure for calibrating the model parameters using experimental stress-strain responses and then demonstrated the sensitivity of deformation to changes in the parameters of the material model and the microstructure [13]. Other methods of mechanical testing have also been utilized to generate experimental responses for the model calibration. For example, Zambaldi et al. used axisymmetric nano-indentation results along with a non-linear optimization procedure to yield a predictive crystal plasticity model where they found a basal to prismatic CRRS ratio of 2.3, and π_1 <c+a> to prismatic CRSS ratio of 7.4 [16]. Although these methods might provide some reasonable comparison with the experimental data; however, their validity strongly depends on the experimental conditions as well as the studied material [12,17]. A short list of CRSS ratios for different slip systems with respect to the CRSS for prismatic slip CRSS is given in Table 1. More comprehensive lists can be found in the literature [7,18].

The short overview given above indicates that the model parameters available in the literature maybe problem-specific and need to be recalibrated to capture experimental results obtained under different conditions, such as different grades of CP-Ti. As pointed out by Wang et al., even for samples with similar chemical compositions and almost the same macroscopic responses, different CRSS values for the crystal plasticity models have been reported [7,14]. This has been postulated to be due to the use of different constitutive frameworks [7], textures, etc. An alternative explanation to this is a possibility that more than one set of crystal plasticity parameters can yield the same macroscopic responses. Additionally, the effects of solutes such as interstitial oxygen further complicate the assessment of CRSS for different slip planes. Typically, in high strength grade Ti alloys, oxygen is intentionally added to the initial amounts already present in the molten metal [1,20], as it improves the strength, corrosion and wear resistance of Ti [21,22]; however, it also enhances strain hardening, which complicates the characterization and simulation of deformation behavior at the macroscopic level [23,24]. Hence, quantifying the CRSS of slip systems as a function of oxygen is important for modeling of macroscopic deformation behavior in Ti alloys. Furthermore, a calibration procedure to obtain optimum model parameters that is less dependent on the setup (or initial conditions) is needed.

In this work, a simple multiscale approach is presented to systematically calibrate a crystal plasticity model applied to an explicit microstructural representation (where individual grains are modeled, i.e., the model is microstructurally explicit) to predict the macroscopic plastic response of pure Ti, and Ti with various dilute oxygen concentrations. Apart from the initial microstructural details such as texture; the proposed crystal plasticity model requires only three key parameters – CRSS ratios for different slip systems, compression and tension strain hardening rates along with elastic constants for model calibration. Fig. 1 below illustrates the general outline for this paper. First, to approximate the CRSS ratios, generalized stacking fault energy (GSFE) calculations through density functional theory (DFT), the Peierls-Nabbaro (PN) model [25], and a continuum-based temperature dependent description of dislocation motion proposed by Kocks et al. [26] were used. Note that thermodynamic variables such as stresses are scale dependent [27]; therefore, it is appropriate to use stress ratios at the continuum scale. Next, tension and compression experiments were performed to obtain the macroscopic response for CP-Ti and CP-Ti with different oxygen concentrations at room temperature along the rolling direction (RD) to determine the strain hardening rates. The calibrated crystal plasticity model using CRSS ratios from lower length scale and slip hardening rates from experiments is shown to be able to replicate the macroscopic response of CP-Ti with different oxygen concentrations along the RD. Then, the crystal plasticity model was used to predict the macroscopic response of CP-Ti with different oxygen concentrations along other loading orientations. Furthermore, the developed macroscopic framework predicted the intrinsic yield behavior[1] of polycrystalline Ti with varying oxygen concentration and the loading-unloading behavior of CP-Ti grade 4 with significant consistency with the available literature data.

2. Experimental methodology

2.1. Material

Rolled plate of CP-Ti grade 2 having a thickness of 1 inch was used to perform mechanical tests and calibrate the crystal plasticity model. The initial interstitial oxygen concentration was 0.16 wt% and the concentration of all other elements were as per the ASTM B265 standard [28]. Microstructure characterization of the as-received CP-Ti was carried out using electron backscatter diffraction techniques (EBSD) as well as X-ray diffraction (XRD). Samples for EBSD were prepared by electropolishing using an A3 electrolyte (Methanol, Ethylene glycol and Perchloric acid) at room temperature and a voltage of 35 V about 12–15 s. EBSD was performed in Zeiss Auriga Dual beam FIB SEM. Scans were carried out at a beam voltage of 20 kV and an aperture size of 120 μm at 70° tilt. EBSD scans were performed on an average area of ~600 μm by ~600 μm with step sizes ranging from 2 to 4 μm. The primary microstructural observations, Fig. 2a, of the as-received CP-Ti sample revealed equiaxed grains with a grain size of 36 μm, averaged over 600 grains, with a standard deviation of 5 μm on the plane perpendicular to the normal direction (ND). EBSD scans were performed at the mid-section of the plate thickness. Pole figure measurements performed using EBSD showed moderate basal texture with the basal pole tilted towards the transverse direction (TD) (see Fig. 2b). Note that additional XRD measurements (not shown here) were carried out on the top and bottom sections of the plate in order to verify the observed texture from EBSD. Fig. 2c illustrates a

[1] The total yield strength is the summation of the intrinsic yield strength and any solid solution strengthening.

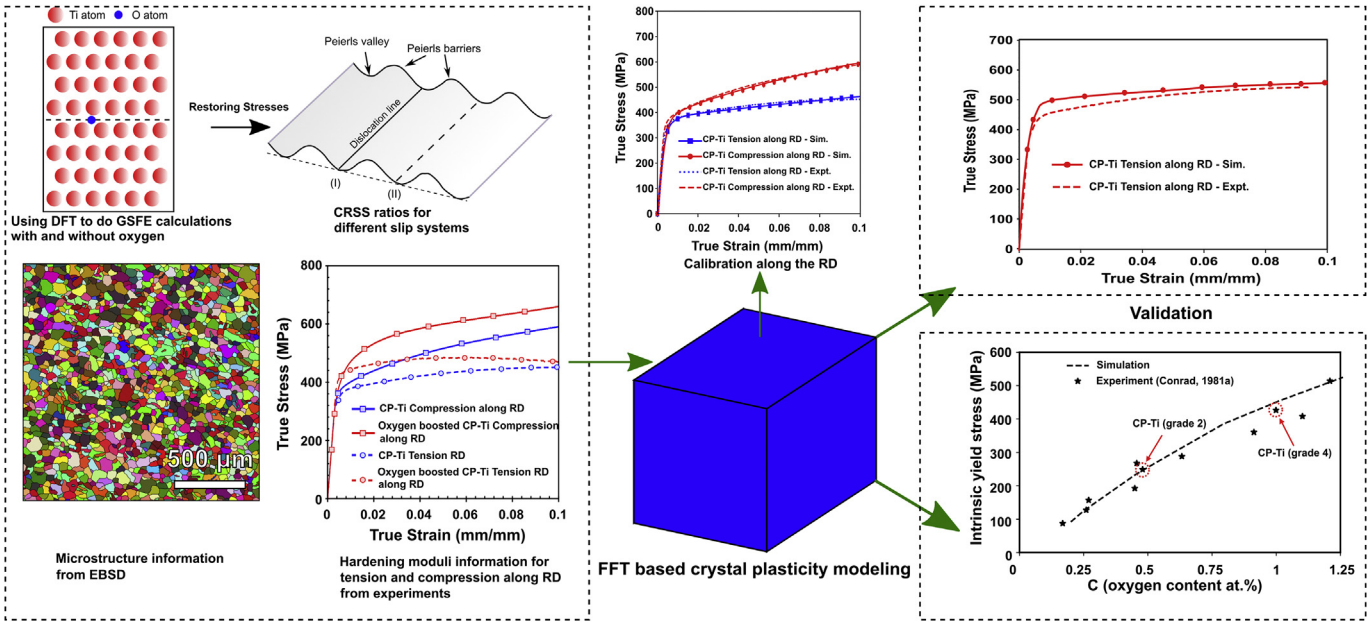

Fig. 1. A general workflow of calibration and validation of the crystal plasticity model to predict the macroscopic plastic response of Ti with different oxygen concentrations. Initial microstructural details and slip hardening rates from experiments along with the lower length scale CRSS ratios were used to calibrate the model. The calibrated crystal plasticity model is able to predict the macroscopic response of CP-Ti for various loading scenarios and also simulate the intrinsic yield behavior of polycrystalline Ti.

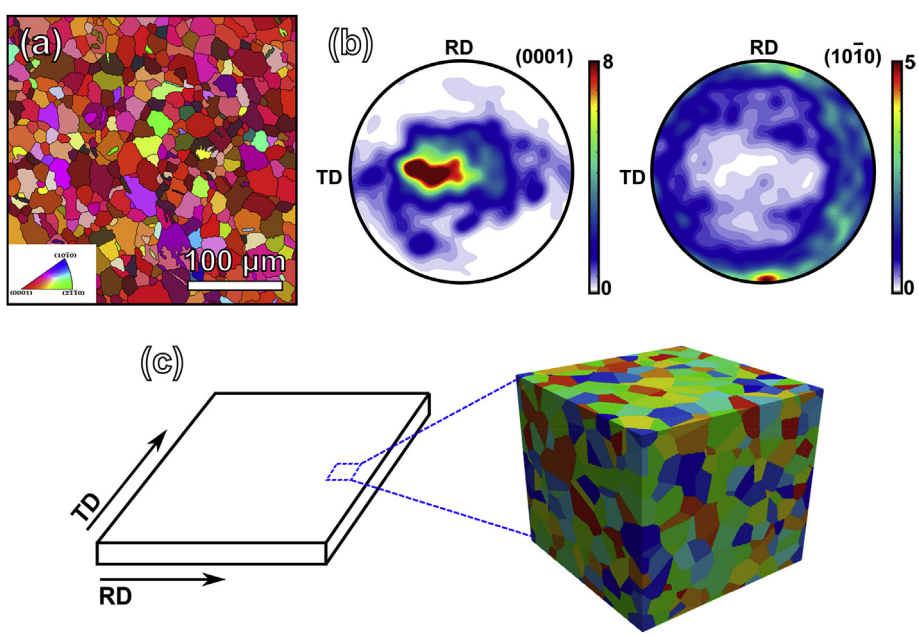

Fig. 2. (a) EBSD scan of CP-Ti taken along the ND. (b) Pole figures generated along the ND from the EBSD scans (c) RVE showing synthetic microstructure with 600 grains used for computational simulations. Experimental data were used to generate the RVE for simulations that represented the experimental microstructure. The crystal orientations were sampled from the experimental orientation distribution and the grains were generated by Voronoi tessellation of the domain based on statistical experimental data (mean grain size and standard deviation) and Python scripts within DAMASK simulation kit.

representative volume element (RVE) for simulations consisting of 600 grains, generated by Voronoi tessellation of the domain based on statistical experimental data (mean grain size and standard deviation) and Python scripts within DAMASK simulation kit. For alloys with microstructure consists equiaxed grains, the Voronoi tessellation has shown to be a reasonable method to approximate the real microstructure [7,29,30]. Further, to guarantee the convergence in all the simulations, 600 grains with a grid of 128x128x128 integration points were used which is significantly larger than various other studies reported in literature, see Refs. [31,32]. Material texture information obtained from measured EBSD data was used to generate an orientation distribution function in order to assign crystal orientations to the grains for the microstructurally-explicit crystal plasticity model (described later) during calibration and validation steps. The assigned texture in the simulation model was compared to the experimental data and was found to be consistent.

2.2. Oxygen boost diffusion heat treatment

Half of the CP-Ti samples were heat-treated in a manner to obtain boost oxygen diffusion from all free surfaces of the cut samples [24]. The heat treatment consisted of two parts, first heating the samples in a carbolated furnace up to 850 °C for 30 min in air to aid oxidation of Ti to TiO_2. Subsequently, the pre-oxidized samples were then heated to 850 °C in ultra-high purity (UHP) Argon atmosphere for 20 h. This enables diffusion of the oxide throughout the thickness from all the free surfaces. The boost oxygen diffusion resulted in a slight gradient of oxygen (even after polishing few microns from each side). Please refer to appendix Figure A1 for the oxygen concentration profile. Henceforth, oxygen boost diffused CP-Ti will be referred as 'oxygen boosted CP-Ti'.

2.2.1. Mechanical testing

Cylindrical specimens (6 mm in length and 6 mm in diameter) were cut along the RD and the ND using electronic discharge machining (EDM) for compression testing. Flat dog-bone specimens were cut using EDM along the RD for tensile testing. Due to thickness limitations along the ND, miniature planar tensile dog-bone specimens (10 mm gauge length and 1 mm thickness) were cut. Specimen geometry for the miniature planar dog-bone specimens can be found in appendix Figure A2. Room temperature uniaxial tension and compression tests were performed at quasi-static strain rates (10^{-3} s^{-1}) using an electromechanical universal testing machine (INSTRON 5900 series) operated with Bluehill software. The experimental stress-strain responses of CP-Ti and oxygen boosted CP-Ti for tension and compression are shown in Fig. 3. Here, the stress-strain responses showed an increase in the yield behavior for different uniaxial loading with increase in the oxygen content. In addition, softening was also observed for the oxygen boosted CP-Ti in tension, which could be attributed to complex interaction between the dislocation and high concentrations of oxygen, and need further advanced characterizations such as in-situ transmission electron microscopy studies [33]. Nonetheless, Fig. 3b shows the EBSD scan of CP-Ti sample that did not reveal noticeable deformation twins after 10% strain along the rolling direction (twin boundaries marked in blue). Similar results were obtained for the oxygen boosted CP-Ti sample (not shown here). Based on the EBSD data, negligible twin fraction was observed post-deformation and hence twinning was not included in the crystal plasticity framework discussed below.

In general, this type of increase in yield and hardening (see Fig. 3a) has been postulated due to strong repulsive interactions between interstitial oxygen and screw dislocation on the prismatic slip system of Ti leading to solid solution strengthening by solute interaction with screw dislocations and also cross-slip [33–37]. However, modeling of such behavior at the continuum scale using crystal plasticity models increases the complexity as well as the number of fitting parameters. In other cases, such as cross-slip, it is nearly impossible to incorporate the exact mechanism. Thus, in order to quantify the effect of oxygen solute atoms on the CRSS ratios for Ti and Ti-O, a set of GSFE (γ-surface as a first approximation) calculations were carried out. These calculations, along with the PN model provide an upper bound (since the explicit modeling of a dislocation core may yield a lower glide stress) to the CRSS ratios and also a constraint to guide the calibration of the crystal plasticity framework proposed here using experimental data. Therefore, we used extensive experimental data to calibrate, and validate the proposed approach.

3. Crystal plasticity model for Ti

A crystal plasticity framework developed by Kalidindi, an extension to the visco-plastic model of Asaro and Needleman, was used to simulate the polycrystalline deformation in Ti [38,39]. Note that the experimental results shown above (Fig. 3b) indicate negligible twin behavior in CP-Ti grade 2 and oxygen boosted CP-Ti, thus, the crystal plasticity framework was restricted to deformation by slip only. The slip systems here are denoted by the unit vectors m^α and n^α, which define the direction and plane normal of the α slip system, respectively. In the following section, the constitutive equations and kinematics of the single crystal plasticity model are briefly presented. The total imposed deformation gradient is multiplicatively decomposed as follows [39]:

$$\mathbf{F} = \mathbf{F}^{*}\mathbf{F}^{p} \tag{1}$$

where $\mathbf{F}^{*}$ and $\mathbf{F}^{p}$ represent elastic (stretch and rotation) and plastic parts of the total deformation gradient, respectively. The time derivative of the plastic (p) part of the deformation can be related to its velocity gradient by Ref. [40].

$$\dot{\mathbf{F}}^{p} = \mathbf{L}^{p}\mathbf{F}^{p} \tag{2}$$

where

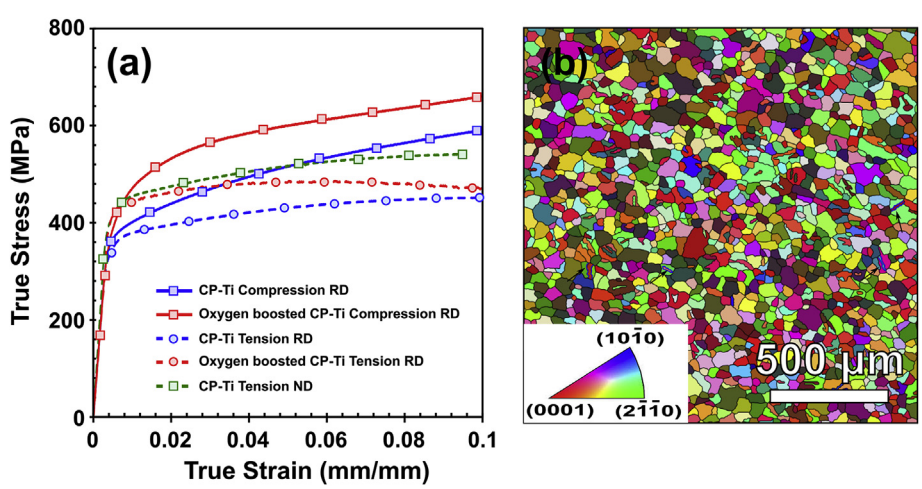

Fig. 3. (a) Experimentally measured stress strain curves along the rolling direction (RD) and the normal direction (ND) for the CP-Ti (grade 2), and oxygen-boosted CP-Ti. (b) post – compression microstructure on the plane perpendicular to the RD for CP-Ti.

$$\mathbf{L}^p = \sum_{\alpha}^{N^{\text{slip}}} \dot{\gamma}^{\alpha} \mathbf{S}_0^{\alpha} \tag{3}$$

$\mathbf{S}_0^{\alpha}$ in the above equation is the Schmid tensor ($\mathbf{m} \otimes \mathbf{n}$) for slip and defined using the initial crystallographic orientation. N^{slip} and $\dot{\gamma}^{\alpha}$ are the total number of slip systems and the slip rate on the αth slip system, respectively. The stress dependence of the slip rate on slip system α is assumed to be governed by the Asaro and Needleman power-law equation [39]:

$$\dot{\gamma}^{\alpha} = \dot{\gamma}_0 \left| \frac{\tau^{\alpha}}{S^{\alpha}} \right|^{\frac{1}{m}} \text{sign}(\tau^{\alpha}) \tag{4}$$

τ and S in Eq. (4) are the resolved shear stress ($\tau = \mathbf{T}^* : \mathbf{S}_0$, where $\mathbf{T}^*$ is the second Piola-Kirchhoff stress tensor) and resistance, respectively, for slip (α) systems. In the above equations, m is the rate sensitivity parameter. More detailed description of the kinematics and the constitutive equations can be found in Refs. [40,41]. Moreover, the evolution of slip hardening is described by the following phenomenological equations [40,41]:

$$\dot{S}^{\alpha} = h_{slip} \left(1 - \frac{S^{\alpha}}{S_{s0}} \right)^{a} \sum_{k}^{N^{slip}} \dot{\gamma}^{k} \tag{5}$$

In the above equations, the material parameter h_{slip}, represent hardening rates (moduli) associated with the slip. S_s^{α} is the overall saturation value of the slip system α, with the twin-free saturation value of S_{s0}. In Eq. (5), the material parameter a is a phenomenological constant and its value was taken from the literature [42,43] (refer to Section 5). Finally, the crystal plasticity model described above was used within the DAMASK fast Fourier transform (FFT) simulation kit [44,45] along with appropriate deformation gradient and complementary stress tensors to simulate simple tension/compression conditions [23]. Note that the model calibration and its predictive nature is discussed in sections 5 and 6.

4. First-principle calculations: role of oxygen on slip resistance

4.1. Generalized stacking fault energies and restoring force calculations at 0 K

The restoring force and mobility of dislocation slip in the lattice can change significantly due to short-range chemical interactions of the matrix with impurity atoms such as oxygen interstitials in Ti [46]. Regarding interstitial oxygen, one explanation is that a normal stress is applied across the dislocation slip plane by the oxygen interstitial, and therefore, the frictional stress of dislocation glide is influenced [23]. This can significantly increase the stress required to move the dislocation. Long-range elastic interactions of the dislocation lines and oxygen interstitial in Ti are relatively weak [47]. Hence, it is postulated that the high strengthening effect of oxygen interstitials comes largely from short-range interactions. First-principles based GSFE calculations have revealed that the presence of oxygen increases unstable stacking fault energy thereby increasing the Peierls stress for dislocation motion on the prismatic slip system of Ti [33,46,48]. First-principles methods were also used to model a screw dislocation dipole on the prismatic slip system of Ti. Strong repulsive interactions were reported between interstitial oxygen and the dislocation core structure leading to solid solution strengthening by solute interaction with screw dislocations [33–37]. However, the presence of solute atoms significantly affects the equilibrium structure of the dislocation core thereby increasing complexity and reducing the accuracy of this approach to determine the stresses required for dislocation glide.

Thus, in this work, in order to quantify the effect of oxygen solute atoms on the frictional stress of dislocation glide on different slip systems of Ti, a set of GSFE (γ-surface) calculations were carried out using the Vienna *ab initio* simulation package (VASP) [49,50]. Projector augmented wave (PAW) [51] pseudopotentials were used to represent the nuclei and core electrons up to the 3p shell with $3d^3$ and $4s^1$ as valence electrons for Ti. Exchange and correlation were treated with the Generalized Gradient Approximation (GGA) using the Perdew-Burke-Ernzerhof (PBE) form [52] with an energy cutoff of 289 eV and the Monkhorst Pack k-point with a gamma mesh of $18 \times 1 \times 10$ for prismatic ($[1\bar{2}10]$, $[10\bar{1}0]$ and $[0001]$), $24 \times 1 \times 12$ for basal-partial ($[1\bar{2}10]$, $[10\bar{1}0]$ and $[0001]$), $12 \times 1 \times 4$ for π_1 plane ($[10\bar{1}2]$,$[10\bar{1}1]$ and $[1\bar{2}10]$) and 24x12x1 for π_2 plane (<c+a>, $[\bar{1}\bar{1}22]$ and $[10\bar{1}0]$) slip systems, respectively. The ionic relaxation was carried out using a conjugate gradient algorithm with 30 meV/Å force and 1 meV energy convergence criteria. The simulation setup for GSFE calculations consisted of 30 atomic layers (2 atoms in each plane) with 10 Å of free space while maintaining periodic boundary conditions. The number of atomic planes and the vacuum region considered were sufficiently large to minimize any interactions between the free surfaces and slip planes. The GSFE curves were calculated by displacing the desired slip plane along the Burger's vector as explained by Bhatia et al. [48,53]. The upper half of the crystal was shifted incrementally with linear steps along the slip direction and allowed to relax at each step. The Ti atoms were allowed to relax only along the direction normal to the slip plane during the GSFE calculations. The oxygen atom was introduced on the shear plane with a monolayer concentration of 1/6 (global concentration ~2 at.%) and was free to relax in all directions during energy minimization of the simulation cell, whereas Ti atoms were free to move along the normal to the slip direction. It is noteworthy that according to the experimental and computational evidence, the <a>-type basal dislocation in HCP materials has a high tendency to dissociate into two $<10\bar{1}0>$ partial dislocations and glide on the basal plane [54]. Moreover, an energetically favorable dissociation of the π_1 <c+a> dislocation into three partials was identified a priori and hence the GSFE calculations were carried out in the direction of the partial dislocation with the highest gradient of energy, which is along the $<10\bar{1}2>$ direction on the π_1 plane, assuming planar glide. For that purpose, the minimum energy path was determined, and a dissociation mechanism was constructed (see Appendix Figure A3 for details).

The PN model describes the dislocation core structure within a continuum framework and can be used to estimate the critical stress required for dislocation motion, i.e., the Peierls stress [55,56]. The restoring force is defined as the gradient of the GSFE in the direction of the minimum energy path and is related to the Peierls stress exponentially [2]. The GSFE and restoring force were obtained for each slip system and are shown in Fig. 4 with and without oxygen. The data with an oxygen atom inserted into the Ti octahedral lattice site is indicated by Ti + O in Fig. 4. The calculated intrinsic stacking fault energy for pure Ti on the prismatic plane is 237 (mJ/m^2), which increases to 276 (mJ/m^2) with insertion of an oxygen atom onto the shear plane. These obtained values compare well with the work of Ghazisaeidi and Trinkle [34,57]. In the case of Ti + O, oxygen increases the barrier for <a> type dislocations on the prismatic plane and decreases the barrier for <c+a> type pyramidal slip systems. Experimental observations indicate an increase in the pyramidal slip activity with oxygen additions [23]. The increase in GSFE on the prismatic slip plane in the case of Ti + O has been correlated with the interstitial volume reduction with oxygen addition, and reported in previous studies on the prismatic slip in Ti

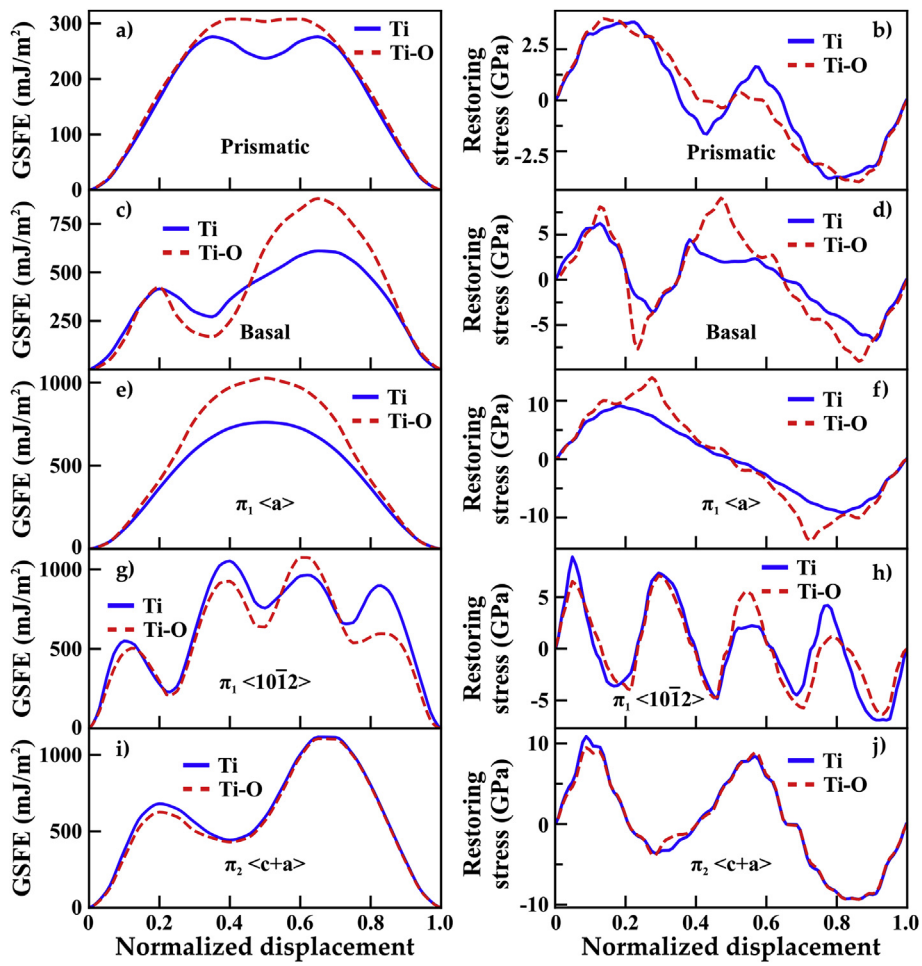

Fig. 4. GSFE (left column) and the corresponding restoring stress (right column) versus the reaction coordinate at 0K for (a) prismatic, (b) basal (partial), (c), π_1 <a>, (d) π_1 <10$\bar{1}$2> (partial) and (e) π_2 <c+a> slip systems in Ti. Horizontal axes show a periodic displacement normalized with respect to the corresponding magnitude of the Burgers vector for each system (refer to the Appendix for the dislocation dissociation reaction on π_1 <c+a>).

[47]. The observed increases in the GSFE on the basal partial and π_1 <a> slip systems of Ti + O further imply the strengthening effects of oxygen interstitials. However, the restoring force on π_1 <10$\bar{1}$2> and π_2 <c+a> slip systems of Ti + O showed a slight decrease compared to that of the pure Ti. Hence, considering the restoring force increase in the other slip systems, it is naturally expected to see more slip activity on π_1 and π_2 planes in Ti samples with more oxygen content (see Fig. 4). The energy barrier calculation with multiple oxygen atoms can lead to periodic image interactions thereby reducing the accuracy of DFT calculations [48].

The prismatic slip system has the lowest unstable stacking fault energy and restoring force both with and without oxygen, and hence, the prismatic slip system can be taken as the most dominant slip system in Ti with various oxygen contents (see Fig. 4a). This observation is consistent with literature reports [34]. Therefore, the obtained Peierls stresses for the other slip systems were normalized with respect to the Peierls stress of the prismatic system. The Peierls stress (σ_p) can be calculated using [58]:

$$\sigma_p = 2K \exp\left(-\frac{4\pi\zeta}{b}\right) \tag{6}$$

where ζ is the dislocation core half width and is defined as [55,58]:

$$\zeta = \frac{Kb}{4\pi F_{\max}} \tag{7}$$

where b is the Burgers vector, K is an energy factor (K is $\mu/(1\text{-}\nu)$ for an edge dislocation and $\mu/2\pi$ for a screw dislocation where μ is shear modulus and ν is the Poisson's ratio) and $F_{\max}$ is the maximum restoring stress. The maximum restoring stress $F_{\max}$ corresponding to a disregistry vector $f(x)$ along the glide plane is defined as [59]:

$$F_{\max} = \left|\frac{d\gamma(f(x))}{df(x)}\right|_{\max} \tag{8}$$

and can be identified from the results in Fig. 4 for each slip system. Intrinsic stacking fault energies (γ_{isf}), unstable stacking fault (γ_{usf}) energies, the ratio $\frac{\gamma_{\text{isf}}}{\gamma_{\text{usf}}}$ and the maximum restoring forces for the studied slip systems are listed in Table 2. The unstable stacking fault energy is associated with a barrier that needs to be overcome to nucleate a partial dislocation [60,61]. The core structure is narrower when $\frac{\gamma_{\text{isf}}}{\gamma_{\text{usf}}}$ is close to unity [55,62]. As seen in Table 2, the addition of oxygen increases the $\frac{\gamma_{\text{isf}}}{\gamma_{\text{usf}}}$ ratio for the prismatic slip. Nevertheless, $F_{\max}$ values for those slip systems are altered considerably (see Table 2). It is noteworthy that the restoring force for π_1 <10$\bar{1}$2> and π_2 <c+a> decreases with the addition of oxygen. The calculated

Table 2
Stacking fault energies, maximum restoring stresses and Peierls stresses (normalized with respect to the prismatic value) for the studied slip families (0K). With the addition of oxygen, the ratio $\frac{\gamma_{isf}}{\gamma_{usf}}$ increases for the prismatic slip system but decreases for the basal partial.

	Slip system (α)	γ_{isf} (mJ/m^2)	γ_{usf} (mJ/m^2)	$\frac{\gamma_{isf}}{\gamma_{usf}}$	F_{max}(GPa)	Peierls stress ratio
Ti	Prismatic	237	276	0.85	3.82	1
	Basal (partial)	271	415	0.65	6.24	2
	π_1 <a>	-	761	-	9.11	6.8
	π_1 <10$\bar{1}$2>	229	549	0.41	8.93	6.6
	π_2 <c+a>	442	680	0.65	10.87	8.6
Ti + O	Prismatic	303	307	0.98	3.99	1
	Basal (partial)	168	424	0.39	8.08	3.5
	π_1 <a>	-	1027	-	13.96	9.7
	π_1 <10$\bar{1}$2>	207	503	0.41	6.51	3.3
	π_2 <c+a>	430	626	0.68	9.47	6.2

Table 3
Non-varying model parameters used in the crystal plasticity simulations. Note that the elastic constants for pure Ti were taken from the literature [69].

Parameter	C_{11} (GPa)	C_{12} (GPa)	C_{13} (GPa)	C_{33} (GPa)	C_{44} (GPa)	$\dot{\gamma}_0$(s^{-1})	m
Value	160	90	66	181.7	46.5	0.001	20

Table 4
The computed CRSS values for pure Ti, commercially pure Ti (CP-Ti, grade 2) and oxygen-boosted CP-Ti. The addition of interstitial oxygen significantly increases the CRSS values for all the slip systems except in the case of π_1 <c+a> slip.

	Slip system (α)	Pure Ti	CP-Ti (grade 2)	Oxygen-boosted CP-Ti
S^{α}(MPa)	Prismatic <a>	37	82	95
	Basal <a>	76	284	328
	π_1 <a>	251	791	915
	π_1 <c+a>	258	268	310
	π_2 <c+a>	320	506	732

restoring forces for different slip systems were then used with Eqs. (6)–(8) to compute the corresponding Peierls stresses for pure Ti and Ti + O, which are listed in Table 2. Here, the Peierls stress for each glide plane was normalized with that of the prismatic plane assuming a screw character (see Table 2). Then, the obtained ratios are used in the Kocks model [26] to compute the CRSS ratios at 300 K in the next section.

4.2. Kocks model for the temperature dependence to compute CRSS ratios at 300 K

Having calculated the absolute zero temperature (0 K) Peierls stresses in the previous section, the room temperature extrapolation is done through a phenomenological model originally proposed by Kocks et al. [26]:

$$\sigma_p(T) = \sigma_p^0 \left\{ 1 - \left(\frac{T}{T_a} \right)^{1/q} \right\}^{1/p} \tag{9}$$

where σ_p^0 is the Peierls stress at absolute zero temperature, T is the temperature, T_a is the temperature threshold for athermal effects, p and q are two material parameters that describe the shape of the temperature dependence of the Peierls stress. T_a is a critical temperature above which the yield stress loses its dependence on the temperature, i.e., the athermal component of the yield stress (intrinsic yield stress) becomes zero [26]. Note that the temperature dependence of individual slip systems can be a complex function of the composition, tendency to cross-slip, dislocation character, etc., and Kocks's formula only provides an estimate of the CRSS value [10,63]. Nevertheless, variations of the thermal component of the yield stress for various polycrystalline as well as single crystal Ti alloys with different oxygen contents have been studied before and a linear dependence with the square root of temperature ($p \approx 1$ and $q \approx 2$) has been reported [10,64,65]. As a first approximation, the same dependence was assumed for all the slip systems here to obtain CRSS values at room temperature. Note that future studies may need to be performed to evaluate p and q values critically for the different slip systems. Nevertheless, the addition of oxygen in Ti induces long, straight and planar screw dislocation arrangements as a dominant deformation mechanism [8,23]. Thus, the CRSS stresses (obtained at 300 K) were then normalized with respect to that of the prismatic system with a screw dislocation character for pure Ti and Ti + O.

Fig. 5 presents the normalized CRSS ratio (room temperature Peierls stresses) estimated by the Kocks model for the studied slip systems of Ti and Ti + O at 300 K. Note that for basal <a> and π_1 <c+a> slip systems, where dislocation dissociation is energetically favorable, it is assumed that the rate controlling resistance is

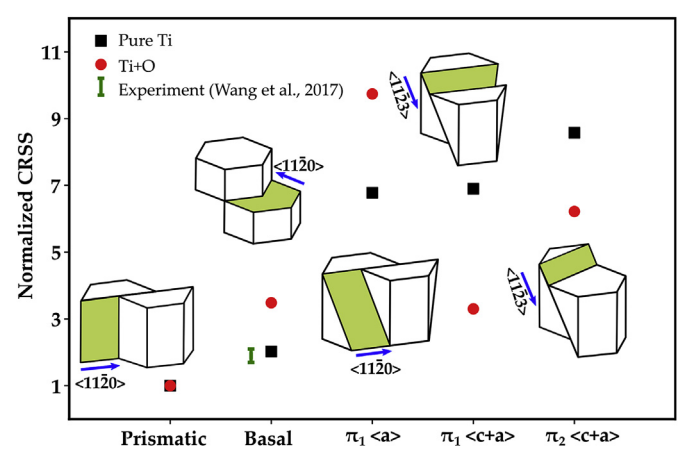

Fig. 5. Room temperature (300K) CRSS ratios for pure and O-doped Ti extrapolated to 300 K using Kocks empirical model from calculations at 0 K using DFT simulations. Values are normalized with respect to the corresponding prismatic slip system in each case. The green bar shows the experimental basal to prismatic CRSS ratio [7]. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

related to the dislocation partial with the highest Peierls stress. The red and black data points in Fig. 5 indicate upper and lower bounds with and without oxygen, respectively, for the normalized CRSS. In other words, the red data points give an estimate of the normalized CRSS ratios assuming the highest oxygen concentration in the octahedral sites on the glide plane. Experimental Ti single crystal and polycrystalline measurements for basal to prismatic CRSS ratios for a variety of oxygen contents (discussed in detail in the introduction) overlap the range of basal to prismatic CRSS values reported in Fig. 5 [7,9]. For example, the basal to prismatic CRSS ratio from the recent experimental work of Wang et al. on polycrystalline Ti is shown in Fig. 5 [7]. In addition, the slip activities of different systems are governed by the ratios of the corresponding CRSS values. In other words, if the CRSS ratios are closer to unity, the plastic deformation is more uniform and vice versa.

Different oxygen contents in Ti can influence the slip activities in a mixed manner and thus it should be taken into account for modeling purposes. For example, it can be seen from Fig. 4 that with the addition of oxygen, the relative resistances of π_1 <c+a> and π_2 <c+a> slip systems decrease, promoting higher slip activity of those systems. This has been reported in some experimental works [23,66,67]. Also, an increase in the relative resistance of the basal slip system to other slip systems translates to less activity of the basal slip system. As mentioned before, Ti alloys are highly anisotropic in the elastic as well as plastic deformation, as clearly indicated by the ratios in Fig. 5. Thus, crystallographic texture plays a crucial role in the macroscopic response of polycrystalline samples. To model such a complicated deformation mechanism at a mesoscale, a systematic calibration of the crystal plasticity model is required not only to reproduce accurately the macroscopic response, but also to be able to study the local microscopic deformation mechanisms and their evolution. In the next section, a crystal plasticity model and its calibration are presented, along with a set of experiments carried out to validate the model.

5. Calibration of the crystal plasticity model: quantifying CRSS of different slip planes

In this section, a model calibration technique is introduced first, which is followed by model validation. The model calibration is critical, especially for HCP materials, due to the strong anisotropy and complexity of the deformation mechanisms that contribute to the large scatter in the reported CRSS values described earlier. The main deformation mechanism in Ti at low strain rates and for various oxygen concentrations is plastic deformation through activation of various slip systems. Therefore, we focus on calibration of the crystal plasticity model for CRSS through a set of high-fidelity crystal plasticity simulations along with the CRSS ratios obtained through lower length scale calculations and a set of tension/compression tests along RD for pure Ti, CP-Ti grade 2 with ~0.16 wt% oxygen, and oxygen-boosted CP-Ti. In case of oxygen-boosted CP-Ti an average ~30% increase in the oxygen content compared to the original CP-Ti samples is observed (see Appendix Figure A1).

The CRSS ratios for pure Ti and Ti with oxygen (Fig. 5) along with the stress-strain curves (Fig. 3) obtained from compression and tension tests along RD were used to calibrate the crystal plasticity model within the DAMASK simulation framework [31,44,45]. Pole figures (see Fig. 2) were used to generate a moderate basal texture and assign the corresponding orientations to 600 grains in the representative volume element (RVE) realizations [68]. The average grain size in the RVE was set to 36 μm in accordance with the experimental microstructure characterization. The non-varying parameters such as elastic constants, etc., used during the calibration process are listed in Table 3 (also refer to Wu et al. [42]). Note that although deformation twin systems in Ti have been reported in the literature, oxygen solute content as well as low strain rates decrease the probability of such a deformation mode [10], also see Fig. 3b which shows negligible twin fraction after deformation. Hence, during the calibration process, the twin-free saturation value (S_{s0}) for all the slip systems was set to be 500 MPa for pure Ti and 1500 MPa for Ti with various interstitial oxygen contents.

Using the present approach, only the slip hardening rate (h_{slip}) and the initial prismatic CRSS were incrementally varied to yield a reasonable agreement between experimentally measured and modelled stress-strain responses. In regard to model parameters, the calibrated CRSS values for the predominant prismatic slip system were found to be 37 MPa, 82 MPa and 95 MPa for pure Ti, CP-Ti grade 2, and oxygen-boosted CP-Ti, respectively (see Table 4). This represents a ~175% increase in the predominant prismatic CRSS value from pure Ti to oxygen-boosted CP-Ti. Note that Table 4 also list calibrated CRSS values for other slip systems, which are calculated based on the CRSS value of the prismatic slip and CRSS ratios computed earlier at 300 K. In regard to the CRSS values of π_1 <c+a> slip, a very small change was found with a varying oxygen content, i.e., CRSS values of 258 MPa, 268 MPa, and 310 MPa for pure Ti, CP-Ti grade 2 and oxygen-boosted CP-Ti, respectively. Interestingly, the addition of interstitial oxygen significantly increases the CRSS values for all of the slip systems except for π_1 <c+a>, which justifies the increased activity of this slip system due to oxygen as reported in the literature [66,70]. In general, both the values and trends of CRSS for the various slip families in Table 4 are within the reported range of the values obtained from polycrystalline and single crystal tension tests; however, a direct comparison is not feasible due to the aforementioned scatter in the reported values [7,10]. Finally, the slip hardening rate (h_{slip}), which was calibrated to be 200 MPa for tension and 650 MPa under compression.

The calibrated stress-strain curves obtained from crystal plasticity model along with its experimental comparison for pure Ti, CP-Ti grade 2 and oxygen-boosted CP-Ti under various stress states along RD are shown in Fig. 6a–c, respectively. The R^2 values of fitted tensile stress-strain curves for both CP-Ti and oxygen-boosted CP-Ti were ~0.98, while that of the compressive stress-strain curves were 0.98 and 0.82 for CP-Ti and oxygen-boosted CP-Ti, respectively. For pure Ti, R^2 value was found to be 0.85. Overall, acceptable calibration results were achieved, since the stress-strain curves obtained from the crystal plasticity model compared well with the experimental responses for pure Ti, CP-Ti grade 2 and oxygen-boosted CP-Ti under different uniaxial loading along the RD. The significant strengthening effect of oxygen content in CP-Ti (0.16 wt%) is apparent from comparing the initial yield stress of pure Ti (~150 MPa) to that of CP-Ti (~375 MPa). The initial yield stress of oxygen-boosted CP-Ti in Fig. 6c shows almost 15% increase from CP-Ti (Fig. 6b). Further, as shown in Fig. 6b and c, the initial yield stress does not noticeably change in compression as compared to tension for the studied variations of oxygen content in Ti. However, the hardening rate increases almost three-fold indicating considerable post-yield tension/compression anisotropy when tested along the rolling direction.

Next, the slip activity, S, for each slip family where plotted (Fig. 7) using following relationship:

$$S = \frac{\sum_{n=1}^{N_s} \gamma_n^S}{\sum_{i=1}^{HCP\ slip\ systems} \sum_{n=1}^{N_i} \gamma_n^i} \tag{10}$$

where N_i is the number of slips within slip system i, was calculated from the calibrated simulations for pure Ti and CP-Ti grade 2 along the RD. As shown in Fig. 7a, the prismatic and basal slip activities are more pronounced in the case of pure Ti. However, as the oxygen

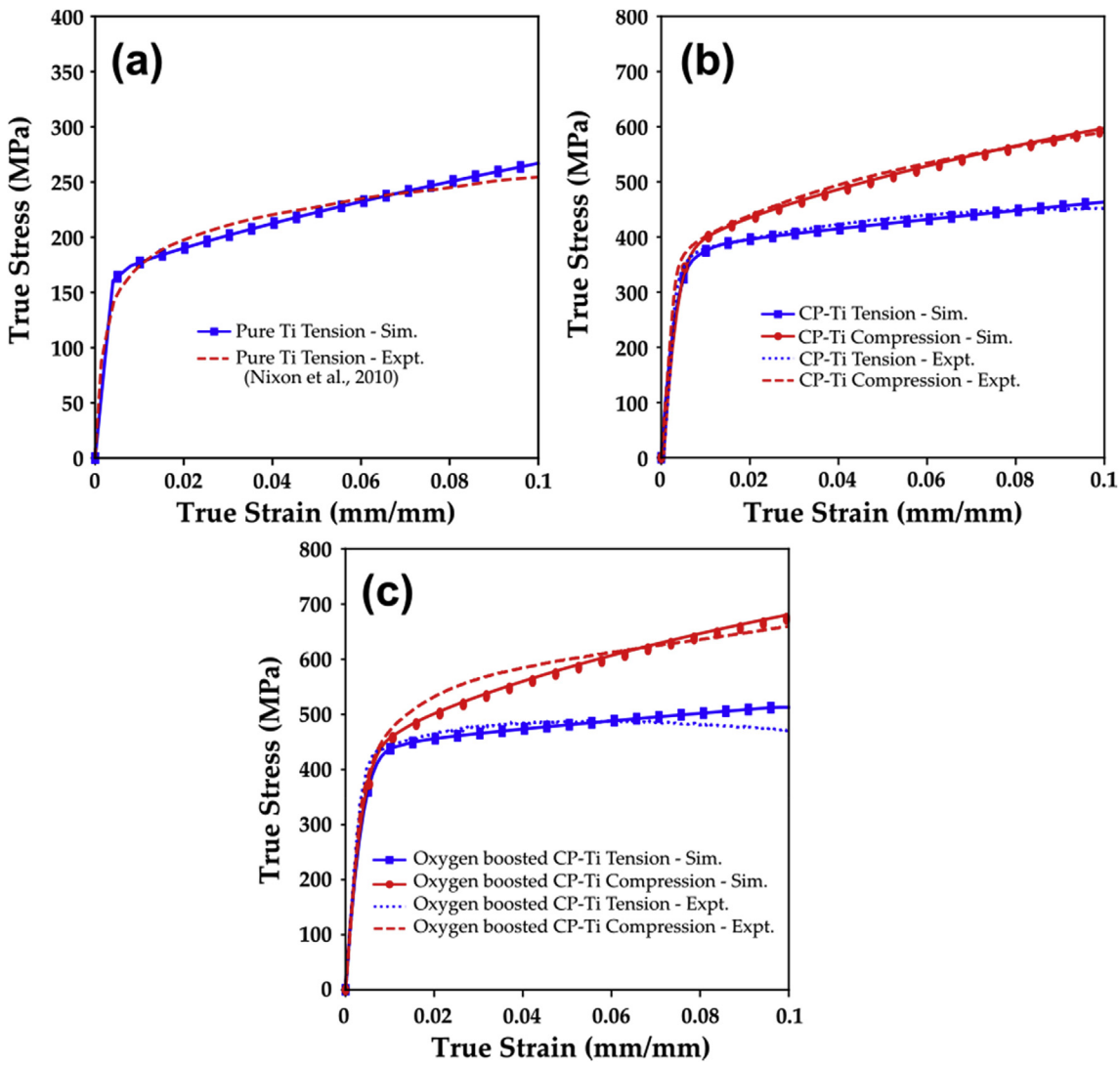

Fig. 6. Model calibration and its comparison with experimentally measured stress strain curves along the rolling direction (RD) for (a) pure Ti [71], (b) CP-Ti (grade 2), and (c) oxygen-boosted CP-Ti.

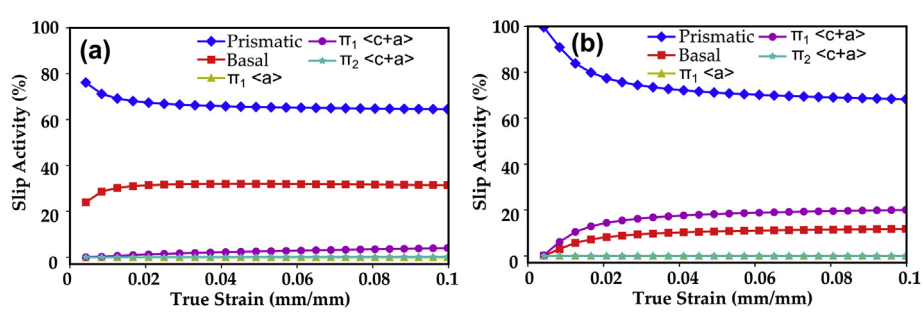

Fig. 7. Simulated slip activities as a function of true strain in (a) pure Ti and (b) CP-Ti grade 2 under tension along the RD.

content increases (Fig. 7b), the activity of π_1 <c+a> increases mostly at the expense of basal slip activity. Further, in both cases, only a small amount of activity of π_1 <a> and π_2 <c+a> slip was observed (see Fig. 7). Interestingly, during the early stage of the plastic deformation, the activity of the prismatic slip system is higher for the case of CP-Ti as compared to pure Ti. However, as the plastic deformation continues, the prismatic slip activity saturates for both cases. In other words, the plastic deformation in pure Ti is mainly controlled by the activation of prismatic and basal slip; however, as the oxygen content increases the deformation is controlled more by prismatic and π_1 <c+a> slip. The reported experimentally measured CRSS values in the literature, where the oxygen effects are not usually stated, may reflect the oxygen content being an uncontrolled variable contributing to the scatter (see Table 1).

6. Validation with a tensile test with a different deformation path

After calibrating the crystal plasticity model along RD, simulations were performed to validate the experimental stress-strain response for CP-Ti grade 2 in tension along the normal direction (ND). Reasonably good qualitative and quantitative agreements ($R^2 = 0.97$) were observed between the experimental and simulated stress-strain curves for the uniaxial tensile loading along ND of CP-Ti grade 2 (Fig. 8a). Note that the crystal plasticity model was calibrated using the stress-strain response along RD only and the stress-strain response along ND was used only for model validation. In addition, the CRSS values and other crystal plasticity model parameters were kept the same during validation as those obtained during the calibration procedure.

Fig. 8a illustrates the stress-strain response along both RD and ND and indicates that the model is capable of capturing the variation in the post yield hardening and flow stress behavior from the rolling to the normal direction. Nevertheless, the predicted stress-strain curve along ND exhibits some divergence from the experimental values, especially immediate post yield behavior. For crystals with their c-axis aligned closer to the loading direction, the tension/compression deformation twin systems are more likely to get activated due to their higher Schmid factor [17,63]. As such, when the samples with moderate basal texture are loaded along the ND, the deformation twin with even a high critical resolved stress can act as a competitive mechanism with the less favorable slip systems. This contributes to the ductility and strain hardening behavior of Ti. Nonetheless, deformation twinning is not considered in the present model, which would explain the higher immediate post yield values in the predicted stress-strain response along ND compared to the experimental values (Fig. 8a). Generally, the normal direction response in the rolled CP-Ti plate has a higher yield value compared to the rolling direction owing to the moderate basal texture. To further understand the underlying plastic deformation activity, the simulated slip activity as a function of true strain is plotted in Fig. 8b. Similar to the rolling direction (Fig. 7b), the prismatic and π_1 <c+a> slip activity is greater for the CP-Ti grade 2. However, the texture utilized here has its c-axis relatively aligned along the normal direction such that the Schmid factor for prismatic slip is on average lower along the ND, and hence, the activity of prismatic slip along ND is lower than in RD, see Figs. 7b and 8b. Interestingly, the slip activity of the π_1 <c+a> system in the CP-Ti grade 2 along the ND is higher compared to that of the RD (compare Figs. 7b and 8b).

To demonstrate broader applicability, the calibrated crystal plasticity model was further validated against a set of experimentally measured intrinsic yield stresses reported by Conrad [10], i.e., the intrinsic yield stress (indicated by σ^*) for different oxygen contents. Previously, the intrinsic yield stress has been reported to show a square root dependence on oxygen concentration ($\sigma^* \propto \sqrt{C}$), also known as Fleischer's theory [72,73]. Thus, we used the calibrated crystal plasticity parameters of CP-Ti grade 2 (an oxygen concentration of ~0.48 at.%) and only scaled the CRSS values for each slip system based on the same square root dependence on oxygen concentrations. A comparison between the experimentally measured intrinsic yield stress and simulated intrinsic yield stress assuming a random texture are given in Fig. 9a, which demonstrates a good qualitative and quantitative agreement. Further, the CRSS values as a function of oxygen content for different slip systems are plotted in Fig. 9b. The prismatic and basal CRSS values at 0.05 at.% of oxygen content in Fig. 9b agree well with the single crystal experimental data reported by Conrad [10], confirming the effectiveness of the present calibration technique and its predictive nature. Overall, Fig. 9b shows CRSS values of five slip systems as a function of oxygen content with the prismatic slip system being the dominant one followed by π_1 <c+a> and basal slip systems.

To further validate the presented approach, the loading-unloading behavior of CP-Ti grade 4 was compared with the available literature data [74], see Fig. 10. The simulated loading and unloading stress-strain curves in Fig. 10 at various strain levels along the transverse direction show a good agreement with the experiments. Here, simulations were performed with CRSS values obtained at 0.96 at.% oxygen from Fig. 9b and the rest of the model parameters remained the same as listed in Table 4. A synthetic texture similar to the experiments was used in the simulations [74]. The reloading yield peaks (dynamic strain aging) in experimental data are mainly due to the oxygen redistribution and accumulation at high dislocation density regions. To accurately model such dynamic strain aging behavior, a multi-physics method to couple the impurity transport and the visco-plastic deformation mechanism would be required. Nevertheless, the calibration technique and predictive nature of the model parameters were validated through simulating mechanical response under various conditions (i.e., stress-state, oxygen contents, etc.), and good agreement between the simulated response and experimental work was found. Further, the multiscale aspect of the calibration technique developed in this work serves as the foundation for future model development and calibration that can be carried out to incorporate the effects of texture and twining.

7. Conclusions

In this work, a crystal plasticity model was calibrated and validated to simulate deformation in polycrystalline Ti on a macroscopic scale with varying oxygen concentration without tuning any fitting parameters once calibration was performed. The microstructurally explicit crystal plasticity model was calibrated using CRSS ratios determined from the first-principle generalized stacking fault energy (GSFE) studies of Ti. The CRSS ratios for Ti and Ti

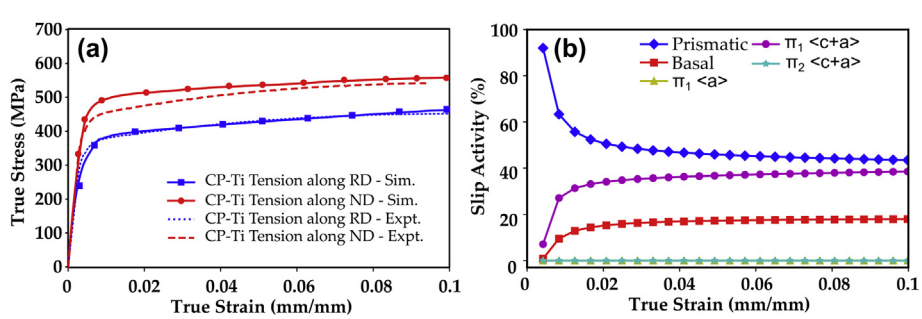

Fig. 8. (a) Stress-strain response of CP-Ti samples along the RD and ND under tension, (b) Simulated slip activity of CP-Ti along the ND under tension resolved on five distinct slip families (compare to Fig. 7b).

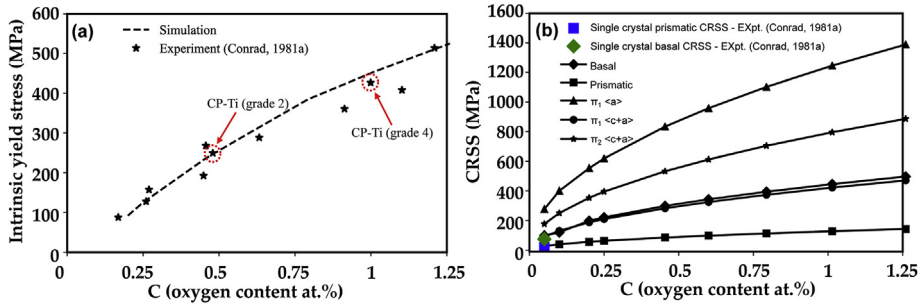

Fig. 9. (a) The simulated square root dependence of the simulated intrinsic yield strength and its comparison with the experimentally measured values [10]. (b) Variation in CRSS values for all the slip systems predicted through crystal plasticity simulations as a function of oxygen content. Note that CRSS ratios obtained earlier were for pure Ti and Ti with highest oxygen concentration in the octahedral sites on the glide plane. Thus, we used the calibrated crystal plasticity parameters of CP-Ti grade 2 (an oxygen concentration of ~0.48 at.%) and only scaled the CRSS values for each slip system based on the same square root dependence on oxygen concentrations.

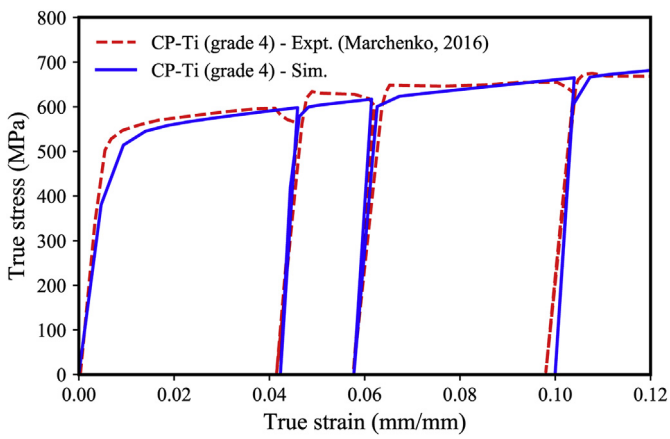

Fig. 10. CP-Ti grade 4 under unloading and reloading quasi-static conditions along the transverse direction (TD) without hold time [74].

with oxygen along with the stress-strain curves obtained from compression and tension tests along the rolling direction were used to calibrate the crystal plasticity model. Further, GSFE calculations were performed to characterize the effect of oxygen on dislocation motion in Ti. The Peierls-Nabbaro (PN) model in conjunction with Kocks model for temperature effects were used to calibrate a visco-plastic crystal plasticity framework to study the effect of oxygen impurity on different slip systems in Ti. The key conclusions from this study are:

1) The calculated GSFE curves for Ti with and without oxygen show that the prismatic is the dominant slip system, followed by basal and π_1 <c+a> slip systems. This is further verified from the simulated slip activities (see Figs. 7 and 8).
2) The CRSS values for the dominant prismatic slip system were found to be 37 MPa, 82 MPa, and 95 MPa for the pure Ti, CP-Ti grade 2 and oxygen-boosted CP-Ti samples computed using the PN model and Kocks model from the GSFE curves.
3) The addition of oxygen increased the energy barrier for all <a>-type dislocations and decreased it for the pyramidal slip systems and, thus, promoted the activity of the π_1 <c+a> slip system.
4) For the experimental cases tested, the calibrated model was proven to be capable of capturing stress-strain response along the ND reasonably well simply by changing the loading direction without a need for modifying any parameters.
5) The average slip activity of the π_1 <c+a> system increased when loading was applied along the ND, indicating the effect of texture on microscopic deformation mechanisms.
6) Simulated intrinsic yield stress and CRSS values as a function of oxygen content demonstrated a good agreement with experimental results.
7) Within its limitations, the predictions by the calibrated crystal plasticity model were found to be in good agreement with the experimental data for Ti having different oxygen concentrations.

Overall, the presented procedure to calibrate the crystal plasticity parameters using the multiscale approach provides a viable pathway for accurate computational material design and can be easily extended to other crystalline materials. The chemical composition and texture properties can significantly influence the macroscopic and microscopic deformation behavior in complex material systems. Therefore, the robust framework presented here can be useful for systematic studies of complicated material/microstructural interactions and enable improved design strategies.

Acknowledgements

The authors gratefully acknowledge support from the Air Force Office of Scientific Research under contract FA9550-13-1-0144.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.actamat.2019.06.040.

Appendix

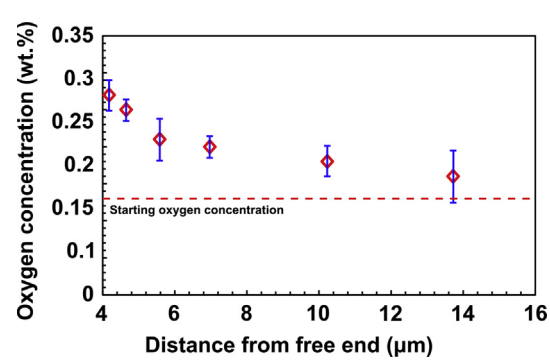

Fig. A1. Electron microprobe analysis (EPMA) of the oxygen boosted CP-Ti sample showing the variation of oxygen as a function of distance from the free surface. Dotted red line indicates the starting oxygen concentration.

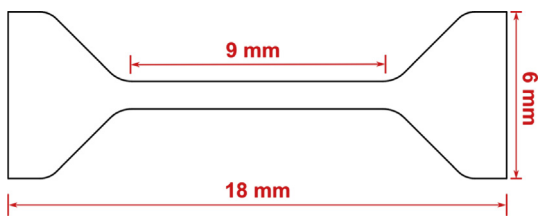

Fig. A2. Miniature dogbone sample used for tensile testing along the ND (thickness of the sample is 1 mm).

a/$311\bar{2}3$, along the identified MEP and along $10\bar{1}2$. As it can be inferred from the figure, energy along the entire $10\bar{1}2$ includes energy along partial directions on the MEP, and that the gradient of energy on those segments is higher leading to higher restoring forces. Hence, DFT calculations for $\pi1$ plane in this paper were only carried out along the whole $10\bar{1}2$ path.

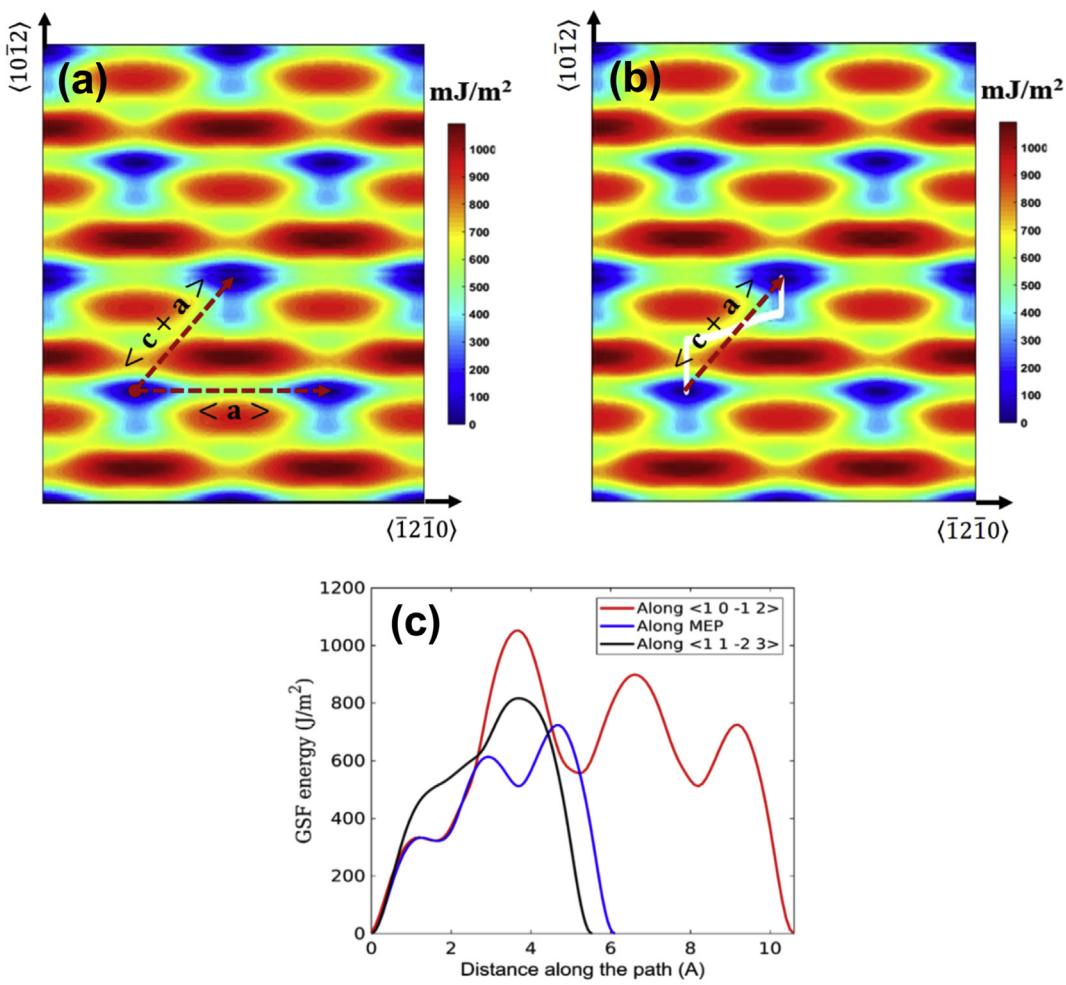

Fig. A3. (a): Generalized stacking fault energy landscape on π_1 plane. Here, x-axis is $\bar{1}2\bar{1}0$ or <a> direction and y-axis is $10\bar{1}2$ direction. Red dashed lines indicate <a> and <c+a> directions, (b) White line shows the calculated MEP consisting of three main segments, (c) GSFE along $10\bar{1}2$, MEP and <c+a> directions.

Generalized stacking fault energy landscape on the first order $\pi1$ plane shown in Figure A3a was calculated using a modified embedded atom method potential for Ti [75]. A complex minimum energy path (MEP) on this plane can be identified. In order to obtain the MEP accurately, the zero-temperature string method was used [76]. A straight line segment connecting two minimum energy points was given as the initial path (a/$311\bar{2}3$) and the segment was iteratively perturbed until the minimum energy path was obtained [76]. The white line in Figure A3b shows the final minimized path which can be described by two vertical segments along $10\bar{1}2$ and one oblique segment along $\bar{1}7\bar{6}5$. Figure A3b depicts energy along

Based on the previous calculations, a dissociation reaction for the π_1 <c+a> slip system can be eventually estimated to be:

$$\frac{a}{3}11\bar{2}3 \;\rightarrow\; a(\alpha 10\bar{1}2 + \beta\bar{1}7\bar{6}5 + \gamma 10\bar{1}2)$$

where $\alpha = 0.2274$, $\beta = 0.0476$ and $\gamma = 0.1535$. The above decomposition satisfies Frank's energy criterion for reaction favorability. Similar dissociations of <c+a> dislocations have been previously reported and associated with tension-compression asymmetry observed in Ti [63].

References

[1] C. Leyens, M. Peters, Titanium and Titanium Alloys: Fundamentals and Applications, John Wiley & Sons, 2003.
[2] P. Garg, M.A. Bhatia, K.N. Solanki, Uncovering the influence of metallic and non-metallic impurities on the ideal shear strength and ductility of Ti: an ab-initio study, J. Alloy. Comp. 788 (2019) 413–421, https://doi.org/10.1016/j.jallcom.2019.02.231.
[3] A.T. Churchman, The slip modes of titanium and the effect of purity on their occurrence during tensile deformation of single crystals, Proc. Roy. Soc. Lond. A. 226 (1954) 216–226, https://doi.org/10.1098/rspa.1954.0250.
[4] F. Barlat, H. Aretz, J.W. Yoon, M.E. Karabin, J.C. Brem, R.E. Dick, Linear transfomation-based anisotropic yield functions, Int. J. Plast. 21 (2005) 1009–1039, https://doi.org/10.1016/j.ijplas.2004.06.004.
[5] T. Hama, A. Kobuki, H. Takuda, Crystal-plasticity finite-element analysis of anisotropic deformation behavior in a commercially pure titanium Grade 1 sheet, Int. J. Plast. 91 (2017) 77–108, https://doi.org/10.1016/j.ijplas.2016.12.005.
[6] A.S. Khan, R. Kazmi, B. Farrokh, Multiaxial and non-proportional loading responses, anisotropy and modeling of Ti–6Al–4V titanium alloy over wide ranges of strain rates and temperatures, Int. J. Plast. 23 (2007) 931–950, https://doi.org/10.1016/j.ijplas.2006.08.006.
[7] L. Wang, Z. Zheng, H. Phukan, P. Kenesei, J.-S. Park, J. Lind, R.M. Suter, T.R. Bieler, Direct measurement of critical resolved shear stress of prismatic and basal slip in polycrystalline Ti using high energy X-ray diffraction microscopy, Acta Mater. 132 (2017) 598–610.
[8] H. Conrad, Effect of interstitial solutes on the strength and ductility of titanium, Prog. Mater. Sci. 26 (1981) 123–403.
[9] B. Barkia, V. Doquet, J.-P. Couzinie, I. Guillot, E. Héripré, In situ monitoring of the deformation mechanisms in titanium with different oxygen contents, Mater. Sci. Eng. A 636 (2015) 91–102, https://doi.org/10.1016/j.msea.2015.03.044.
[10] H. Conrad, Effect of interstitial solutes on the strength and ductility of titanium, Prog. Mater. Sci. 26 (1981) 123–403, https://doi.org/10.1016/0079-6425(81)90001-3.
[11] A.S. Khan, Y. Sung Suh, R. Kazmi, Quasi-static and dynamic loading responses and constitutive modeling of titanium alloys, Int. J. Plast. 20 (2004) 2233–2248, https://doi.org/10.1016/j.ijplas.2003.06.005.
[12] N. Benmhenni, S. Bouvier, R. Brenner, T. Chauveau, B. Bacroix, Micromechanical modelling of monotonic loading of CP α-Ti: correlation between macroscopic and microscopic behaviour, Mater. Sci. Eng. A 573 (2013) 222–233, https://doi.org/10.1016/j.msea.2013.02.022.
[13] J.R. Mayeur, D.L. McDowell, A three-dimensional crystal plasticity model for duplex Ti–6Al–4V, Int. J. Plast. 23 (2007) 1457–1485, https://doi.org/10.1016/j.ijplas.2006.11.006.
[14] A.A. Salem, S.R. Kalidindi, S.L. Semiatin, Strain hardening due to deformation twinning in α-titanium: constitutive relations and crystal-plasticity modeling, Acta Mater. 53 (2005) 3495–3502, https://doi.org/10.1016/j.actamat.2005.04.014.
[15] B.D. Smith, D.S. Shih, D.L. McDowell, Cyclic plasticity experiments and polycrystal plasticity modeling of three distinct Ti alloy microstructures, Int. J. Plast. 101 (2018) 1–23, https://doi.org/10.1016/j.ijplas.2013.10.004.
[16] C. Zambaldi, Y. Yang, T.R. Bieler, D. Raabe, Orientation informed nanoindentation of α-titanium, J. Mater. Res., Journal of Materials Research 27 (2012) 356–367, https://doi.org/10.1557/jmr.2011.334.
[17] J.S. Weaver, M.W. Priddy, D.L. McDowell, S.R. Kalidindi, On capturing the grain-scale elastic and plastic anisotropy of alpha-Ti with spherical nanoindentation and electron back-scattered diffraction, Acta Mater. 117 (2016) 23–34, https://doi.org/10.1016/j.actamat.2016.06.053.
[18] H. Li, D.E. Mason, T.R. Bieler, C.J. Boehlert, M.A. Crimp, Methodology for estimating the critical resolved shear stress ratios of α-phase Ti using EBSD-based trace analysis, Acta Mater. 61 (2013) 7555–7567, https://doi.org/10.1016/j.actamat.2013.08.042.
[19] J. Gong, A.J. Wilkinson, Anisotropy in the plastic flow properties of single-crystal α titanium determined from micro-cantilever beams, Acta Mater. 57 (2009) 5693–5705, https://doi.org/10.1016/j.actamat.2009.07.064.
[20] ASM Handbook, Volume 2: Properties and Selection: Nonferrous Alloys and Special-Purpose Materials - ASM International, ASM International, 1990.
[21] J.R. Davis, Properties and Selection: Nonferrous Alloys and Special-Purpose Materials, Asm Intl, 1990.
[22] J. Stringer, The oxidation of titanium in oxygen at high temperatures, Acta Metall. 8 (1960) 758–766, https://doi.org/10.1016/0001-6160(60)90170-X.
[23] M.C. Brandes, M. Baughman, M.J. Mills, J.C. Williams, The effect of oxygen and stress state on the yield behavior of commercially pure titanium, Mater. Sci. Eng. A 551 (2012) 13–18, https://doi.org/10.1016/j.msea.2012.04.058.
[24] H. Dong, X.Y. Li, Oxygen boost diffusion for the deep-case hardening of titanium alloys, Mater. Sci. Eng. A 280 (2000) 303–310, https://doi.org/10.1016/S0921-5093(99)00697-8.
[25] F.R.N. Nabarro, Fifty-year study of the Peierls-Nabarro stress, ResearchGate 234 (1997), https://doi.org/10.1016/S0921-5093(97)00184-6.
[26] U.F. Kocks, A.S. Argon, M.F. Ashby, Thermodynamics and Kinetics of Slip, Pergamon Press, 1975.
[27] K. Solanki, M. Horstemeyer, M. Baskes, H. Fang, Multiscale study of dynamic void collapse in single crystals, Mech. Mater. 37 (2005) 317–330.
[28] B10 Committee, Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate, ASTM International, n.d. doi:10.1520/B0265-15.
[29] R. Quey, P.R. Dawson, F. Barbe, Large-scale 3D random polycrystals for the finite element method: generation, meshing and remeshing, Comput. Methods Appl. Mech. Eng. 200 (2011) 1729–1745, https://doi.org/10.1016/j.cma.2011.01.002.
[30] S. Bargmann, B. Klusemann, J. Markmann, J.E. Schnabel, K. Schneider, C. Soyarslan, J. Wilmers, Generation of 3D representative volume elements for heterogeneous materials: a review, Prog. Mater. Sci. 96 (2018) 322–384, https://doi.org/10.1016/j.pmatsci.2018.02.003.
[31] F. Roters, M. Diehl, P. Shanthraj, P. Eisenlohr, C. Reuber, S.L. Wong, T. Maiti, A. Ebrahimi, T. Hochrainer, H.-O. Fabritius, S. Nikolov, M. Friák, N. Fujita, N. Grilli, K.G.F. Janssens, N. Jia, P.J.J. Kok, D. Ma, F. Meier, E. Werner, M. Stricker, D. Weygand, D. Raabe, DAMASK – the Düsseldorf Advanced Material Simulation Kit for modeling multi-physics crystal plasticity, thermal, and damage phenomena from the single crystal up to the component scale, Comput. Mater. Sci. 158 (2019) 420–478, https://doi.org/10.1016/j.commatsci.2018.04.030.
[32] F. Bridier, D.L. McDowell, P. Villechaise, J. Mendez, Crystal plasticity modeling of slip activity in Ti–6Al–4V under high cycle fatigue loading, Int. J. Plast. 25 (2009) 1066–1082, https://doi.org/10.1016/j.ijplas.2008.08.004.
[33] Q. Yu, L. Qi, T. Tsuru, R. Traylor, D. Rugg, J.W. Morris, M. Asta, D.C. Chrzan, A.M. Minor, Origin of dramatic oxygen solute strengthening effect in titanium, Science 347 (2015) 635, https://doi.org/10.1126/science.1260485.
[34] M. Ghazisaeidi, D.R. Trinkle, Interaction of oxygen interstitials with lattice faults in Ti, Acta Mater. 76 (2014) 82–86, https://doi.org/10.1016/j.actamat.2014.05.025.
[35] M. Ghazisaeidi, D.R. Trinkle, Core structure of a screw dislocation in Ti from density functional theory and classical potentials, Acta Mater. 60 (2012) 1287–1292, https://doi.org/10.1016/j.actamat.2011.11.024.
[36] N. Tarrat, M. Benoit, D. Caillard, L. Ventelon, N. Combe, J. Morillo, Screw dislocation in hcp Ti : DFT dislocation excess energies and metastable core structures, Model. Simul. Mater. Sci. Eng. 22 (2014) 055016, https://doi.org/10.1088/0965-0393/22/5/055016.
[37] N. Tarrat, M. Benoit, J. Morillo, Core structure of screw dislocations in hcp Ti: an ab initio DFT study, IJMR 100 (2009) 329–332, https://doi.org/10.3139/146.110055.
[38] S.R. Kalidindi, Incorporation of deformation twinning in crystal plasticity models, J. Mech. Phys. Solids 46 (1998) 267–290, https://doi.org/10.1016/S0022-5096(97)00051-3.
[39] R.J. Asaro, A. Needleman, Overview no. 42 Texture development and strain hardening in rate dependent polycrystals, Acta Metall. 33 (1985) 923–953, https://doi.org/10.1016/0001-6160(85)90188-9.
[40] M. Ardeljan, R.J. McCabe, I.J. Beyerlein, M. Knezevic, Explicit incorporation of deformation twins into crystal plasticity finite element models, Comput. Methods Appl. Mech. Eng. 295 (2015) 396–413, https://doi.org/10.1016/j.cma.2015.07.003.
[41] M. Knezevic, H.F. Al-Harbi, S.R. Kalidindi, Crystal plasticity simulations using discrete Fourier transforms, Acta Mater. 57 (2009) 1777–1784, https://doi.org/10.1016/j.actamat.2008.12.017.
[42] X. Wu, S.R. Kalidindi, C. Necker, A.A. Salem, Prediction of crystallographic texture evolution and anisotropic stress–strain curves during large plastic strains in high purity α-titanium using a Taylor-type crystal plasticity model, Acta Mater. 55 (2007) 423–432, https://doi.org/10.1016/j.actamat.2006.08.034.
[43] S.R. Kalidindi, Modeling anisotropic strain hardening and deformation textures in low stacking fault energy fcc metals, Int. J. Plast. 17 (2001) 837–860, https://doi.org/10.1016/S0749-6419(00)00071-1.
[44] P. Eisenlohr, M. Diehl, R.A. Lebensohn, F. Roters, A spectral method solution to crystal elasto-viscoplasticity at finite strains, Int. J. Plast. 46 (2013) 37–53, https://doi.org/10.1016/j.ijplas.2012.09.012.
[45] F. Roters, P. Eisenlohr, L. Hantcherli, D.D. Tjahjanto, T.R. Bieler, D. Raabe, Overview of constitutive laws, kinematics, homogenization and multiscale methods in crystal plasticity finite-element modeling: theory, experiments, applications, Acta Mater. 58 (2010) 1152–1211.
[46] J.A. Yasi, L.G. Hector Jr., D.R. Trinkle, First-principles data for solid-solution strengthening of magnesium: from geometry and chemistry to properties, Acta Mater. 58 (2010) 5704–5713, https://doi.org/10.1016/j.actamat.2010.06.045.
[47] Q. Yu, L. Qi, T. Tsuru, R. Traylor, D. Rugg, J.W. Morris, M. Asta, D.C. Chrzan, A.M. Minor, Origin of dramatic oxygen solute strengthening effect in titanium, Science 347 (2015) 635–639, https://doi.org/10.1126/science.1260485.
[48] M.A. Bhatia, X. Zhang, M. Azarnoush, G. Lu, K.N. Solanki, Effects of oxygen on prismatic faults in α-Ti: a combined quantum mechanics/molecular mechanics study, Scripta Mater. 98 (2015) 32–35, https://doi.org/10.1016/j.scriptamat.2014.11.008.
[49] P. Garg, I. Adlakha, K.N. Solanki, Effect of solutes on ideal shear resistance and electronic properties of magnesium: a first-principles study, Acta Mater. 153 (2018) 327–335, https://doi.org/10.1016/j.actamat.2018.05.014.
[50] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, Comput. Mater. Sci. 6 (1996) 15–50, https://doi.org/10.1016/0927-0256(96)00008-0.
[51] P.E. Blöchl, Projector augmented-wave method, Phys. Rev. B 50 (1994) 17953–17979, https://doi.org/10.1103/PhysRevB.50.17953.
[52] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized gradient approximation made

simple, Phys. Rev. Lett. 77 (1996) 3865, https://doi.org/10.1103/PhysRevLett.78.1396. Phys. Rev. Lett. 78 (1997) 1396–1396.
[53] M.A. Bhatia, M. Rajagopalan, K.A. Darling, M.A. Tschopp, K.N. Solanki, The role of Ta on twinnability in nanocrystalline Cu–Ta alloys, Mater. Res. Lett. 5 (2017) 48–54, https://doi.org/10.1080/21663831.2016.1201160.
[54] Z. Wu, M.F. Francis, W.A. Curtin, Magnesium interatomic potential for simulating plasticity and fracture phenomena, Model. Simul. Mater. Sci. Eng. 23 (2015) 015004, https://doi.org/10.1088/0965-0393/23/1/015004.
[55] M.A. Bhatia, M. Azarnoush, I. Adlakha, G. Lu, K.N. Solanki, Dislocation core properties of β -tin: a first-principles study, Model. Simul. Mater. Sci. Eng. 25 (2017) 025014, https://doi.org/10.1088/1361-651X/aa57d0.
[56] V. Vitek, Dislocation cores and unconventional properties of plastic behavior, in: S. Yip (Ed.), Handbook of Materials Modeling: Methods, Springer Netherlands, Dordrecht, 2005, pp. 2883–2896, https://doi.org/10.1007/978-1-4020-3286-8_174.
[57] M. Ghazisaeidi, D.R. Trinkle, Core structure of a screw dislocation in Ti from density functional theory and classical potentials, Acta Mater. 60 (2012) 1287–1292, https://doi.org/10.1016/j.actamat.2011.11.024.
[58] G. Lu, N. Kioussis, V.V. Bulatov, E. Kaxiras, Generalized-stacking-fault energy surface and dislocation properties of aluminum, Phys. Rev. B 62 (2000) 3099–3108, https://doi.org/10.1103/PhysRevB.62.3099.
[59] H. Mori, H. Kimizuka, S. Ogata, Dislocation properties and Peierls stress of BCC iron based on generalized-stacking-fault energy surface by using first principles calculations, J. Jpn. Inst. Metals 73 (2009) 595–600, https://doi.org/10.2320/jinstmet.73.595.
[60] M.A. Bhatia, I. Adlakha, G. Lu, K.N. Solanki, Generalized stacking fault energies and slip in β-tin, Scripta Mater. 123 (2016) 21–25, https://doi.org/10.1016/j.scriptamat.2016.05.038.
[61] J.R. Rice, Dislocation nucleation from a crack tip: an analysis based on the Peierls concept, J. Mech. Phys. Solids 40 (1992) 239–271, https://doi.org/10.1016/S0022-5096(05)80012-2.
[62] H. Van Swygenhoven, P.M. Derlet, A.G. Frøseth, Stacking fault energies and slip in nanocrystalline metals, Nat. Mater. 3 (2004) 399–403, https://doi.org/10.1038/nmat1136.
[63] I.P. Jones, W.B. Hutchinson, Stress-state dependence of slip in Titanium-6Al-4V and other H.C.P. metals, Acta Metall. 29 (1981) 951–968, https://doi.org/10.1016/0001-6160(81)90049-3.
[64] A. Akhtar, E. Teghtsoonian, Prismatic slip in α-titanium single crystals, Metall. Mater. Trans. A 6 (1975) 2201, https://doi.org/10.1007/BF02818644.
[65] T. Tanaka, H. Conrad, Deformation kinetics for {1010}⟨1120⟩ slip in titanium single crystals below 0.4Tm, Acta Metall. 20 (1972) 1019–1029, https://doi.org/10.1016/0001-6160(72)90136-8.
[66] F. Bridier, P. Villechaise, J. Mendez, Analysis of the different slip systems activated by tension in a α/β titanium alloy in relation with local crystallographic orientation, Acta Mater. 53 (2005) 555–567, https://doi.org/10.1016/j.actamat.2004.09.040.
[67] Z. Liu, The Effects of Oxygen and Heat Treatment on Microstructures and Mechanical Properties of Alpha and Beta Titanium Alloys, Ph.D., Case, Western Reserve University, 1988, https://search-proquest-com.ezproxy1.lib.asu.edu/docview/303654450/abstract/F5ABCC1DCE8E4CF1PQ/1. (Accessed 2 March 2018).
[68] R. Hielscher, H. Schaeben, A novel pole figure inversion method: specification of the MTEX algorithm, J Appl Cryst, J Appl Crystallogr. 41 (2008) 1024–1037, https://doi.org/10.1107/S0021889808030112.
[69] D. Tromans, Elastic anisotropy of HCP metal crystals and polycrystals, Int. J. Res. Rev. Appl. Sci. 6 (2011) 462–483.
[70] S. Naka, A. Lasalmonie, P. Costa, L.P. Kubin, The low-temperature plastic deformation of α-titanium and the core structure of a-type screw dislocations, Philos. Mag. A 57 (1988) 717–740, https://doi.org/10.1080/01418618808209916.
[71] M.E. Nixon, O. Cazacu, R.A. Lebensohn, Anisotropic response of high-purity α-titanium: experimental characterization and constitutive modeling, Int. J. Plast. 26 (2010) 516–532, https://doi.org/10.1016/j.ijplas.2009.08.007.
[72] R. Fleischer, The Strengthening of Metals, Reinhold, New York, 1964, p. 93.
[73] R. Fleischer, W. Hibbard Jr., Relation between Structure and Strength in Metals and Alloys, 1963.
[74] A. Marchenko, M. Mazière, S. Forest, J.-L. Strudel, Crystal plasticity simulation of strain aging phenomena in α-titanium at room temperature, Int. J. Plast. 85 (2016) 1–33, https://doi.org/10.1016/j.ijplas.2016.05.007.
[75] M.I. Baskes, R.A. Johnson, Modified embedded atom potentials for HCP metals, Model. Simul. Mater. Sci. Eng. 2 (1994) 147, https://doi.org/10.1088/0965-0393/2/1/011.
[76] E. Weinan, R. Weiqing, V.-E. Eric, Simplified and improved string method for computing the minimum energy paths in barrier-crossing events, J. Chem. Phys. 126 (2007) 164103, https://doi.org/10.1063/1.2720838.